



Mustafa Ramadan | Research Assistant

I plan to be a part of ground-breaking research study in the field of Quantum Information and Technology. With the critical thinking, thirst for knowledge and computer skills that I've built through my work and volunteer experiences, I will be able to pursue that goal and achieve it.

And this is merely a step in my life journey. My mission is to keep adding upon the knowledge tower that humans have built by learning and doing research.

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EXPERIENCE

April 2024 –
April 2025

University of Cologne

| Master Thesis

With the supervision of PD. Dr. Klesse, I'm working on the Quantum Channel capacity Problem. I'm starting with analytical calculation of the **purity coherent information** for n shots. And working on the numerical calculation of higher n .

November 2022 –
May 2023

University of Cologne

| Research Assistant

During this project, I've been using **C++** to merge two data structures to better **visualize scientific data** and being able to render the results using blender and other commercial tools. The goal is to **create a poster** out of this research in order to entice more future research about this topic.

April 2020 –
September 2021

Study Smart Series

| Customer service

My duties started as customer support, but I immediately saw a chance to improve upon the initiative and implemented the structure of the course in an automated fashion using **Java-script** and Telegram bot API which allows the teacher to focus his/her time on creative ideas, rather than doing repetitive tasks.

August 2019 –
September 2020

Tafaouky Center

| Manager/R&D

The center's goal was to focus on helping graduate students in a seamless transition between university and their dream career by providing the right training for them.

My role was mainly to improve upon the quality of the training by researching the required skills for each career line and implementing it in the curriculum.

March 2017 –
January 2020

Reaction Team

| Teaching Assistant

By mapping and planning lessons, I was able to be a part of a team that provided higher quality education for high school and university students.

September 2018 –
January 2019

Ebad Al-Rahman School

| Teacher

Using active learning I was able to teach physics to more than 300 middle school students and show its importance in our daily life.

February 2016 –
September 2018

Link Team for Development

| Volunteer/Manager

Building and managing a team that introduces physics to more than 1000 children and to more than 3000 high school graduates

August 2015 –
June 2017

Plus Library

| Fact-Checker & Editor

We aimed to explain the lectures given by the professor in a simple language. We did so by reading about the content of each lecture from multiple textbooks and then summarizing everything in a cohesive written format.

I was responsible for fact-checking: **E&M**, Series and Complex Analysis, introduction to solid-state, **Analytical Mechanics** and **Numerical Analysis & Programming**.

And Summarizing: Series and Differentiation, Vector Analysis, Scientific English, **Quantum Physics** and Special Functions.



EDUCATION

October 2021
- June 2025

MSc in Physics

| Universität zu Köln

In this program, I focused on **quantum information** and quantum technologies, where I have learned the fundamentals of that field that would allow me to keep up with current research. I also gained a lot of experience in **programming** and **data analysis** through **python** courses and lab experiments.

September 2021

Bachelor in Physics

| Damascus University

During my bachelor, we learn a lot of knowledge about the laws that governs our universe, but the most important thing that I got out is problem solving and critical thinking skills.

2019

Understanding Research Methods

| Coursera | University of London

During this active course I was able to learn more about research planning and the skills required for the researchers. I was fortunate to use these skills during my work at Tafaouky Center.

2018

Fundamentals Skills for Volunteers

| UNFPA & SEBC

The skills I got from this course helped me better form my understanding of management and I was able to deploy those skills at Link Team for development.

2017

Public Speackign Art

| ARSCP

Delivering ideas wasn't my strong suit, but the skills acquired from this course helped me excel at my teaching assistant job at Reaction team.

Late 2016

TESOL: Teaching English for Speakers of Other Languages

| Business & TESOL Academy

Teaching methods is a very powerful tool that should be an essential skill to any researcher, and that's exactly what I got out of this wonderful and coherent course.



LANGUAGES

English: TOEFL IBT Test - C1

Reading

Listening

Writing

Speaking

German: University Course -B1

Reading

Listening

Writing

Speaking



SKILLS

Computer Skills

Python

Matlab

Latex

C++

Soft Skills

Problem Solving

Commu nication

initiative

Research

Ort, Datum

Unterschrift